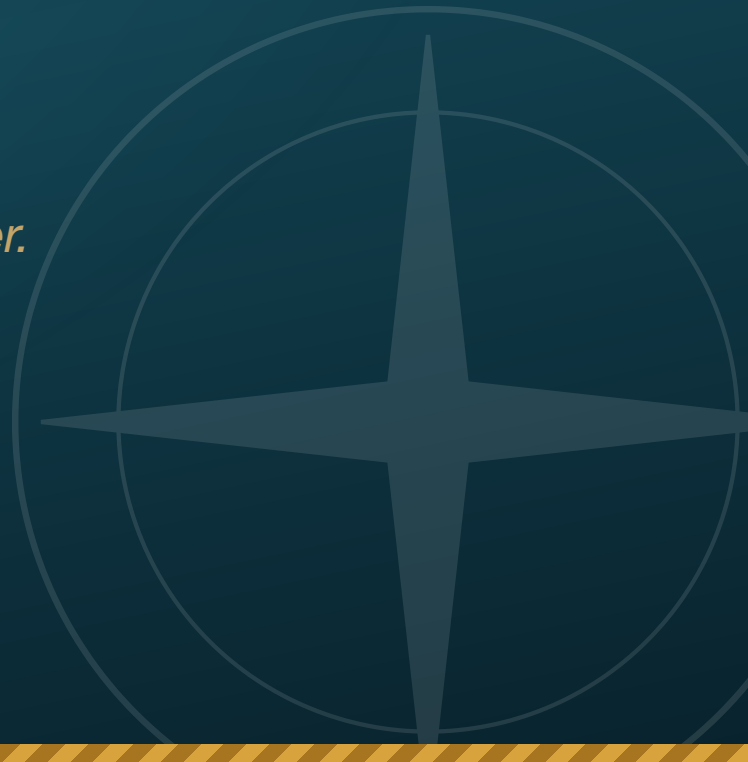
DAY 1 · LECTURE 2 · 45 MINUTES

Into the Deep

From Containers to Kubernetes Pods

Your raft is built. Now we launch it into open water.

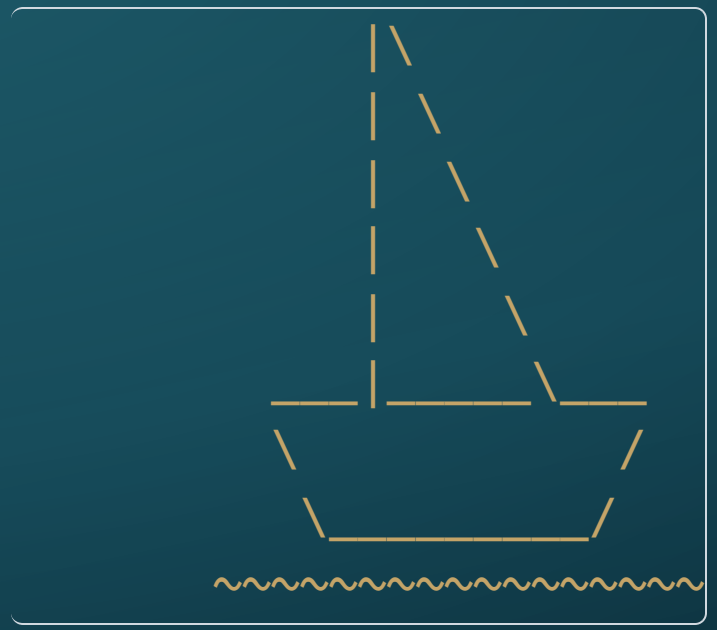


Where we are

- Lab 01 is done — you have a container **image** sitting in Harbor.
- That's a raft, built and beached. It doesn't sail itself.
- The next 45 minutes: how Kubernetes takes that image and puts it to sea.
- Four short legs:
  - Why one container isn't a fleet
  - The **Pod** — what Kubernetes actually runs
  - Declaring what you want, instead of hand-steering
  - Why Pods are built to be thrown away
- Then a quick flash poll, and **Lab 02 — Paddling Out**.

PART I

Why a Raft Isn't a Fleet



> *"One raft is a tale. A fleet is a future. Ye can't crew the whole ocean alone." — the Boatswain*

One container, one problem

- On your VM you can already do this: `docker run my-raft`
- It works — until it doesn't:
  - the container **crashes** and stays down
  - the **host reboots** and nothing comes back
  - you need **ten** copies, not one
  - the machine it runs on fails, and it has to move
- A lone container has nothing watching it. Something has to — and "a person doing it by hand" is not a plan that survives contact with reality.

You need a harbour-master

- Something that watches every container and **keeps the fleet sailing**:
  - restarts the ones that sink
  - schedules them across whatever machines you have
  - scales them up and down on demand
  - replaces them when a whole machine is lost
- That something is an **orchestrator**.

Meet the orchestrator: Kubernetes

- **Kubernetes** is the orchestrator the island runs on — the book calls it **Captain Kube**.
- You hand Captain Kube your containers and a description of what you want.
- It makes the ocean **match that description** — and keeps it matching.
- *We meet Captain Kube properly tomorrow morning.* Today: just enough to paddle out.

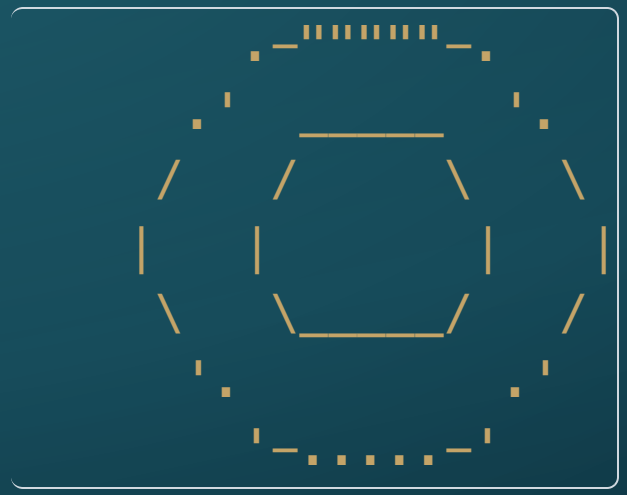
The real shift: telling vs. asking

- **Docker** is *imperative* — "run this container, right now." A one-time order.
- **Kubernetes** is *declarative* — "I want one of these to **always** exist."
- You describe the **destination**. The cluster does the steering, continuously.
- Pull a Pod out from under it, and Kubernetes notices and acts.

> *Stop hand-steering every wave. Hand over the chart and let the crew hold course.*

PART II

A Life-Jacket Called a Pod



> *"Never send a crate to sea without a life-jacket. The deep don't forgive." — the Boatswain*

Kubernetes doesn't run containers

- Here's the surprise: the smallest thing Kubernetes runs is **not** a container.
- It's a **Pod**.
- A raw container is just a box. Kubernetes wraps that box in a life-jacket — and the life-jacket is what it actually schedules, watches, and restarts.

So what is a Pod?

- A **Pod** is the smallest deployable unit in Kubernetes.
- It wraps **one — or more — containers** that are always:
  - **scheduled together**, on the same machine
  - **started and stopped together**, sharing one lifecycle
  - **networked together**, sharing one address
- Most Pods hold exactly **one** container. But the wrapper can hold more — and that turns out to be useful.

Why wrap it? Containers in a Pod share

- One **IP address** for the whole Pod — containers inside reach each other on `localhost`.
- Shared **storage** — they can hand files back and forth.
- One **lifecycle** — born together, die together.

The sidecar pattern

- Most Pods are one container. But you can add a **helper** alongside it:
  - a **log shipper** that forwards your app's logs
  - a **proxy** that handles encryption or routing
  - a **file-syncer** that pulls in fresh content
- Main container + sidecar, sharing one Pod's network and storage.
- The app stays simple; the helper does the plumbing — **without changing your image**.

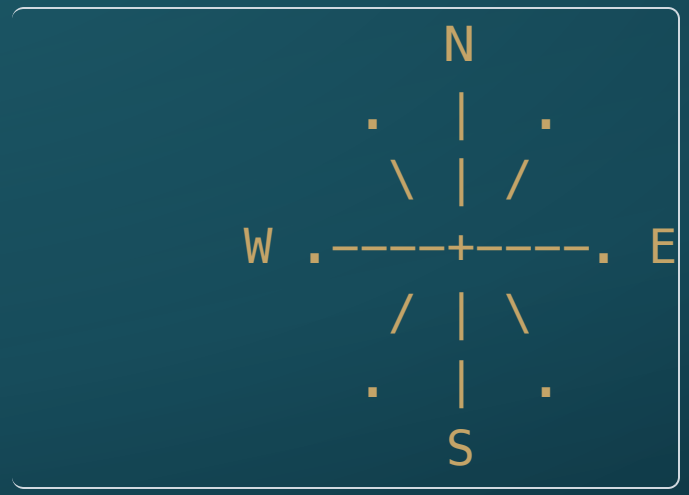
Pod vs. Container

	Container	Pod
What it is	a packaged process	Kubernetes' smallest unit
Holds	one application	one *or more* containers
Network	(on its own)	one shared IP address
You...	**build** it (Lab 01)	**deploy** it (Lab 02)

You build containers. You deploy Pods. The Pod is the box Kubernetes can actually carry.

PART III

Charts, Not Hand-Steering



> *"Hand the sea a good chart and it'll hold your course while ye sleep." — the Boatswain*

Kubernetes objects are written down

- Everything in Kubernetes — a Pod included — is described in a **manifest**: a **YAML** file.
- YAML is just structured text: keys, values, indentation.
- A Pod manifest can run 30+ lines, and it **looks** intimidating.
- Good news, and the rule for this whole seminar:

> **You do not write Kubernetes YAML from memory. You generate it.**

Generate, don't memorise

```
kubectl run my-raft \
  --image=harbor.wagbiz.org/raft-fleet/<your-
name>:v1 \
  --dry-run=client -o yaml > pod.yaml
```

- `--dry-run=client` — *"build the manifest, but don't create anything yet."*
- `-o yaml` — print it as YAML.
- `> pod.yaml` — save it to a file you can open and read.
- Kubernetes just wrote your manifest **for** you.

The workflow: generate, read, apply

```
1. GENERATE          2. READ / EDIT          3. APPLY
-----------          --------------          -----------
kubectl run     -->  open pod.yaml      --> kubectl apply
--dry-run=client     plain, readable         -f pod.yaml
-o yaml > pod.yaml   YAML you can tweak       => Pod is live
```

- You never face a blank file. You start from a working draft.
- Read it, adjust it if you need to, then **apply** it to the cluster.

Claim your territory: Namespaces

- One cluster. Around **thirty** sailors sharing it.
- A **Namespace** is your own patch of ocean — your work, walled off from everyone else's.

```
kubectl create namespace <your-name>
kubectl config set-context --current --namespace=<your-name>
```

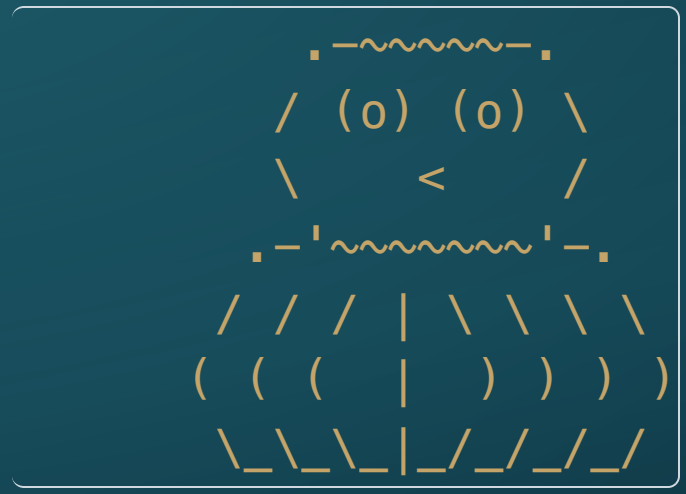
- The second line says "*always work in my patch*" — so you stop typing `-n` every time.
- Your `my-raft` Pod and your neighbour's never collide.

Seeing the whole ocean

- `kubectl get pods` — the Pods in **your** namespace.
- `kubectl get pods -A` — **every** Pod, in **every** namespace, across the whole cluster.
- That `-A` flag is how you see past your own patch of water.
- Keep it in your back pocket — Lab 02 ends with a reason to need it.

PART IV

Ghost Ships



> *"You best start believing in ghost stories, Miss Turner... you're in one." — Captain Barbossa, Pirates of the Caribbean: The Curse of the Black Pearl*

Cattle, not pets

- A Pod is **disposable** — by design.
- It can crash, be rescheduled to another machine, or be replaced at any moment.
- That is not a failure. That is Kubernetes **keeping the chart matched**.
- So: don't name it, don't nurse it, don't hand-tune it. **Don't grow attached.**

The Ghost Ship

```
kubectl exec into a running Pod
      |
      +--  hand-edit a file inside it      (looks fine!)
      |
kubectl delete pod  +  kubectl apply -f pod.yaml
      |
      v
the hand-edit is GONE - rebuilt from the immutable image
```

- Changes you make **inside** a running Pod live only in that Pod.
- Delete it, recreate it from `pod.yaml`, and those changes **vanish**.

> *If it isn't baked into the* **image** *or written in the* **manifest**, *it will sink.*

Your diagnostic spyglass

When a Pod misbehaves, three commands tell you almost everything:

```
kubectl get pods                    is it running?
what's its status?
kubectl logs <pod>                  what did the
container print?
kubectl exec -it <pod> -- sh  board the Pod and
look around
```

- `get` for the **status**, `logs` for the **story**, `exec` to **walk the deck**.
- You'll use all three in Lab 02 — and lean on them all week.

Instructor Superpower

ONE CLUSTER, THIRTY SAFE SANDBOXES

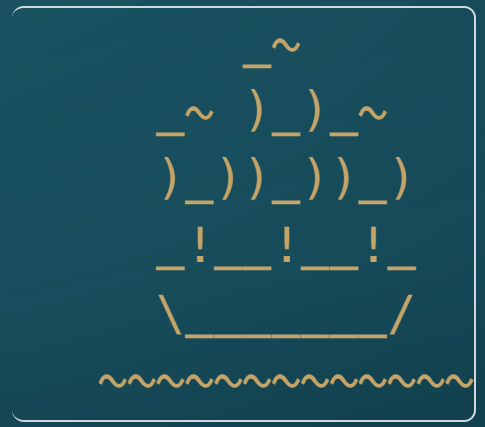
- Every student's work is two things you can **see**: a readable **manifest** and an isolated **namespace**.
- A student stuck? `delete` the Pod, re-`apply` the file — a clean reset in seconds.
- The real shift: the environments you can *choose* to support have radically expanded. You don't wait for IT to say "sure, we can support that" — you build it yourself, for a lesson, a project, a whole course.

Up next: Flash Poll, then Lab 02

- **Flash poll** — five quick questions to check the crew before we sail. Open `poll.wagbiz.org`.
- Then **Lab 02 — Paddling Out**:
  - board the live cluster with your Rancher kubeconfig
  - claim your **namespace**
  - **generate** `pod.yaml`, then `apply` it
  - practice the spyglass: `logs` and `exec`

> *Outcome: your raft afloat in the live cluster — and you, able to tell why when it isn't.*

Paddle out.



The raft is yours. The ocean is the cluster. Go.